Cutnell & Johnson
Newton's Third Law of Motion Part 2
Tug-of-War
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

Date:	
Your Name:	
Group:	
Period:	

Introduction *Exploration Goals*

§ The purpose of this exploration is to determine the forces exerted on two objects in a tug-of-war. You will measure and compare the forces on each of the two bodies.

§ For this activity, use Force Sensors attached to carts to measure the force of interaction during a tug-of-war between the two carts.

Learning Outcomes

You will be able to:

• Compare the force exerted on one cart during a tug-of-war to the force exerted on the second cart during the tug-of-war.

• Use a graph of force versus time for each cart to determine the force exerted on each cart during a tug-of-war.

• Describe Newton's Third Law of Motion in conceptual terms.

• Demonstrate a quantitative understanding of Newon's Third Law by applying it to a problem related to a tug-of-war.

Hypothesis

§ How does the force on one end of the rope in a traditional tug-of-war compare to the force on the other end of the rope (assuming that the rope is in static equilibrium)?

§ How do the forces measured by two Force Sensors pulling away from each other compare? Are they equal?

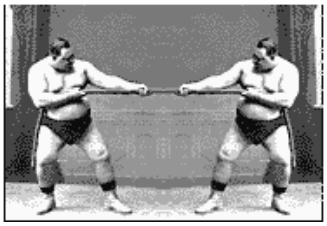
Background

§ Whenever one body exerts a force on a second body, the second body exerts an oppositely directed force of equal magnitude on the first body.

§ The third law is often called the “action-reaction” law, for it is sometimes quoted as follows: “For every action (force) there is an equal, but opposite, reaction.”

§ In a traditional tug-of-war, the winning team is usually the one that has the best traction against the ground. The forces exerted by each team pulling on the rope are equal in magnitude and opposite in direction.

» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 4, Section 4.5.

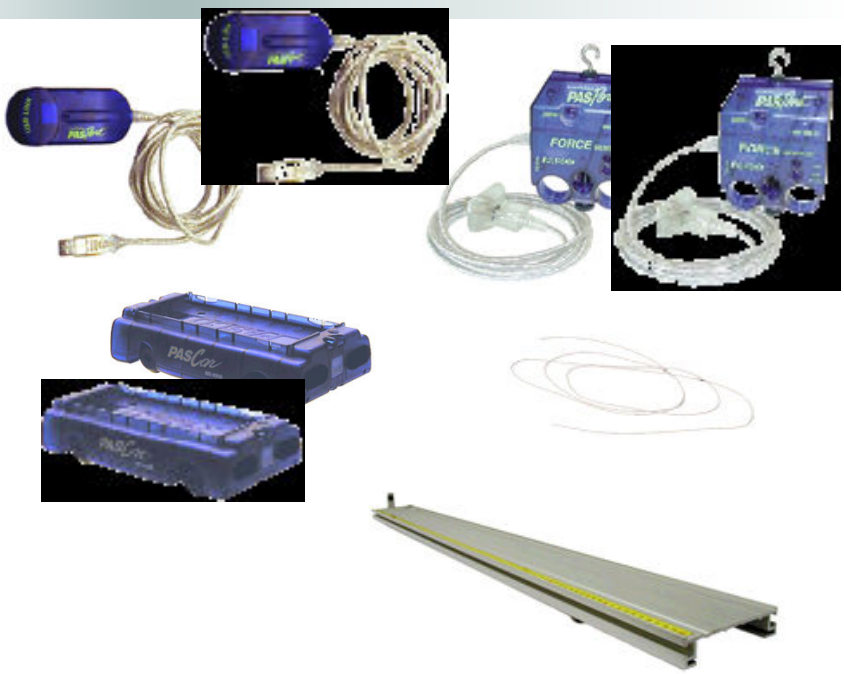


$$F_{12} = -F_{21}$$

Newton's Third Law of Motion

Materials

- 2 PASPORT USB Links
- 2 PASPORT Force Sensors
- 2 PAScars
- String
- 1.2 m Dynamics Track



Setup

Equipment Setup

Safety Reminder: Follow the directions for using the equipment.

§ Place the Dynamics Track on a horizontal surface. Level the Dynamics Track by placing the PAScar on the Dynamics Track. If the cart rolls one way or the other, use the Adjustable Foot at one end of the Dynamics Track to raise or lower that end until the Dynamics Track is level and the cart does not roll one way or the other.

§ Use the thumbscrew that comes with the Force Sensor to mount the sensor onto the accessory tray of each cart.

§ Put a hook on the front of each sensor. Place the carts on the track so the hooks face one another.

§ Tie a loop of string around the hooks on the Force sensors. The string should be about 20 cm (8 inches).

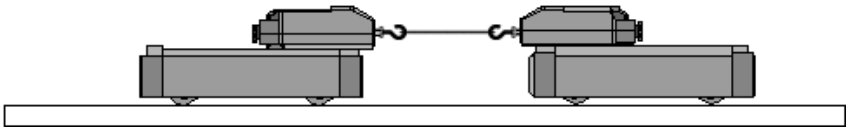
Note: It is very important that the track is level to get the best results.

Process *Prepare to Record Data*

§ It is important to **Zero** the Force Sensors prior to each data run.

» The idea is to pull the two carts apart as if in a tug-of-war. The Force Sensors will measure the forces exerted on each cart during the pulling.

» The graph of force versus time is set up so that the measurements from the two sensors will be plotted on opposite sides of the X-axis. (One sensor is "Force, pull right" and the other sensor is "Force, pull left".) The area under each curve corresponds to the force.



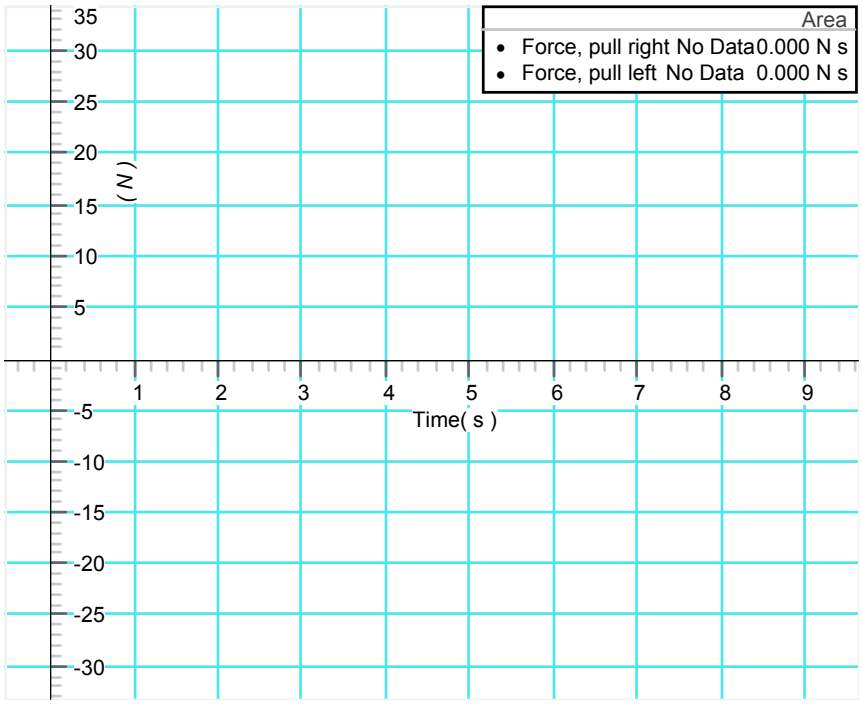
Process

Record Data

§ **Zero** both sensors and click **Start**. Slowly pull the carts apart. Do not exceed 50 N or you could ruin the sensors.

§ Slowly release the carts and click **Stop**.

» The next page shows how to analyze the force versus time data.



Analyze *How to find the Force*

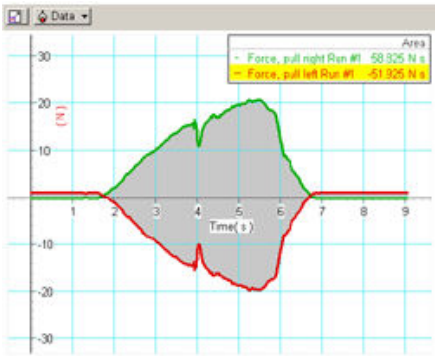
§ Click **Scale to Fit** if needed.

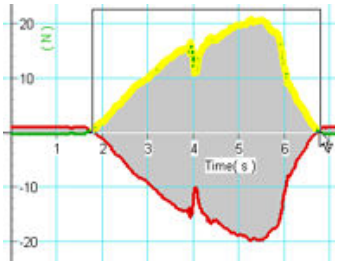
§ Highlight the top curve as in the example. The Legend box shows the Area under the curve for data from the first sensor.

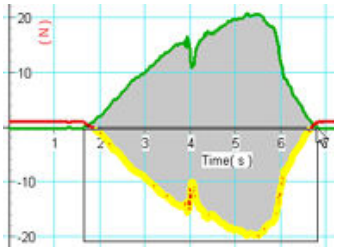
§ Highlight the bottom curve as in the example. The Legend box shows the Area under the curve for data from the second sensor.

§ Record the values for the Area for each sensor on the next page.

Click Scale to Fit if needed

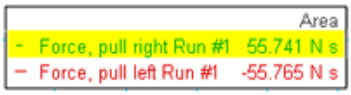






Select regions of the curves

Legend box



Analyze

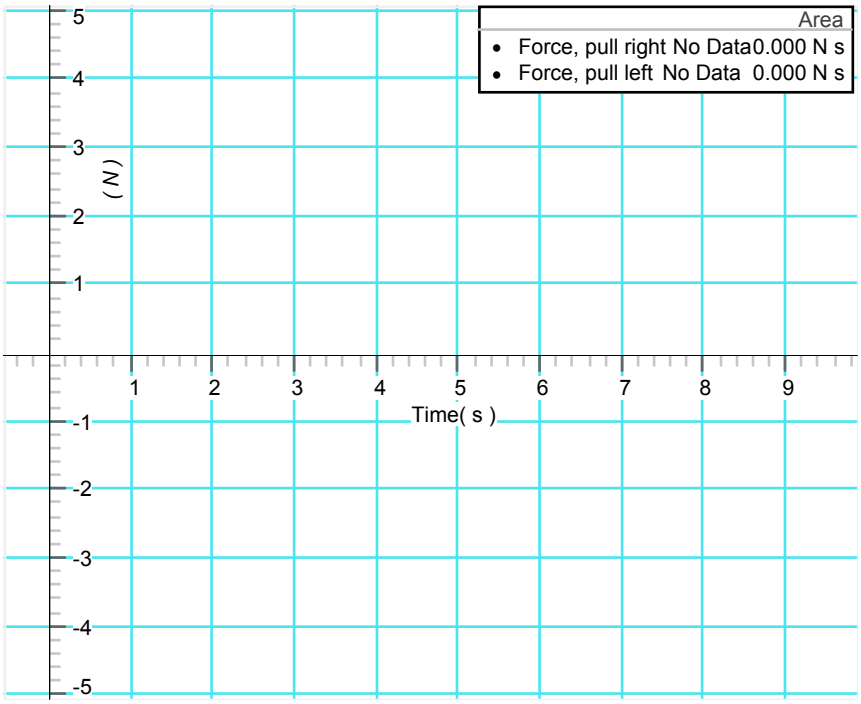
Find the Force

§ Use the procedure described previously to find the Area under the curve for both sensors.

§ Record the values on this page.

Force, pull right: Area

Force, pull left: Area



Observations

§ How did the shape of the graph for the first Force Sensor compare to the shape of the graph for the second Force Sensor during the Tug-of-War?

Synthesize *Comparison and Error Analysis*

§ How does the force measured by the first Force Sensor compare to the force measured by the second Force Sensor?

§ What were the sources of error in this experiment?

Force, pull right: Area

Force, pull left: Area

Synthesize *Conclusions*

§ During a tug-of-war, how does the force on one Force Sensor compare to the force on the other Force Sensor?

§ How does your hypothesis compare to your results? Hypothesis:

Synthesize *Applications*

§ What are examples of physical events where Newton's Third Law applies?

Synthesize *Extension Problem*

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 4, problem 16, page 117.

At a time when mining asteroids has become feasible, astronauts have connected a line between their 3500-kg space tug and a 6200-kg asteroid. Using their ship's engine, they pull on the asteroid with a force of 490 N. Initially the tug and the asteroid are at rest, 450 m apart. How much time does it take for the ship and the asteroid to meet?

Notes